

LOCAL LLM QUESTION-ANSWERING SYSTEM

Question 3 – Implementation

1. Question

Integrate a locally running Large Language Model with a Python application to implement a question-answering system.

2. Aim

To develop an interactive question-answering application using Python, Streamlit, Ollama, and a locally running Llama 3.2 Large Language Model. The system accepts a user's question and generates a clear answer locally.

3. Technologies Used

- Python 3.12.10
- Streamlit
- Ollama 0.33.2
- Llama 3.2 – local Large Language Model
- VS Code

4. Project Structure

Local_LLM_Text_Generation/

- |— venv/ – Python virtual environment
- |— app.py – Question 1 text generation
- |— summarization.py – Question 2 text summarization
- |— qa.py – Question 3 question answering

5. Implementation Steps

1. Use the existing Local_LLM_Text_Generation project folder and virtual environment.
2. Activate the existing virtual environment using `.\venv\Scripts\Activate.ps1`.
3. Reuse the installed Streamlit and Ollama Python packages.
4. Reuse the locally installed Llama 3.2 model.
5. Create a new Python file named `qa.py`.
6. Create a Streamlit interface containing a local model selector and question input area.
7. Construct a prompt that instructs the LLM to answer the user's question clearly and accurately.
8. Send the question to Llama 3.2 through the Ollama Python library.
9. Receive the generated answer and display it in the Streamlit application.
10. Run the application using `streamlit run qa.py`.

6. Python Implementation

```
import streamlit as st
import ollama

st.set_page_config(
    page_title="Local LLM Question Answering",
    page_icon=" ? ",
    layout="centered"
)

st.title(" ? Local LLM Question Answering")

st.write(
    "Ask questions and receive answers from a locally running "
    "Large Language Model through Ollama."
)

model = st.selectbox(
    "Select Local LLM",
    ["llama3.2"]
)

question = st.text_area(
    "Enter your question",
    placeholder="Example: What is Artificial Intelligence?",
    height=150
)

if st.button("Get Answer", type="primary", width="stretch"):

    if not question.strip():
        st.warning("Please enter a question.")

    else:
        prompt = f"""
You are a helpful question-answering assistant.

Answer the user's question clearly and accurately.
Use simple language and provide enough explanation
to understand the answer.

Question:
{question}

Answer:
"""

        response = ollama.chat(
            model=model,
            messages=[
                {"role": "user", "content": prompt}
```

```
]
)

answer = response["message"]["content"]

st.subheader("Answer")
st.write(answer)
```

7. Working Principle

The user enters a question through the Streamlit interface. The Python application places the question into a structured prompt and sends it to Ollama. Ollama runs the Llama 3.2 model locally, which generates the answer. The response is returned to Python and displayed in the Streamlit interface.

8. System Workflow

11. User enters a question.
12. Streamlit captures the question.
13. Python creates the question-answering prompt.
14. The prompt is sent to Ollama.
15. Llama 3.2 processes the question locally.
16. The model generates an answer.
17. The answer is returned to Python.
18. Streamlit displays the generated answer.

9. Input

Question: What is Artificial Intelligence?

10. Output

The local Llama 3.2 model generated an explanation of Artificial Intelligence, including its purpose and examples of tasks such as understanding data, recognizing patterns, making decisions and learning from experience.

11. Application Output Screenshot

? Local LLM Question Answering

Ask questions and receive answers from a locally running Large Language Model through Ollama.

Select Local LLM

llama3.2

Enter your question

What is Artificial Intelligence?

Get Answer

Answer

Artificial Intelligence (AI) is a type of computer technology that allows machines to think and learn like humans. It's a way to create intelligent machines that can perform tasks that typically require human intelligence, such as:

- Understanding and interpreting data
- Recognizing patterns and images
- Making decisions and predictions
- Learning from experience

AI systems use complex algorithms and large amounts of data to make decisions and take actions. They

Figure 1: Streamlit interface showing local LLM question answering

12. Result

Thus, a Python-based question-answering system was successfully integrated with a locally running Llama 3.2 Large Language Model through Ollama. The system accepts user questions and displays generated answers through an interactive Streamlit interface.

13. Conclusion

The implementation demonstrates the integration of a local Large Language Model with a Python application for question answering. Since the model runs through Ollama on the local system, the application can process questions without requiring a cloud-based LLM API.